

resistor R2, the base of transistor T1 remains high and it does not conduct. As a result, the voltage at pin 3 of IC CA3130 (IC1) remains low. The potential divider comprising resistors R4 and R5 maintains half of 5.1V at pin 2 of IC1. In brief, the voltage at pin 2 of IC1 is higher than at pin 3 and its output remains low. LED2 remains 'off' and transistor T2 does not conduct. Relay RL1 remains de-energised and, as a result, security lamps (both indoors and outdoors) remain switched off.

When you press any key of the remote TV handset, IR rays fall on the receiver (IRX1) and its output goes low. LED1 flashes in sync with pulsation of the IR rays. At the same time, transistor T1 (BC558) conducts to take pin 3 of IC1 high. IC1 is used as a comparator with timer action.

When transistor T1 conducts, pin 3 of IC1 gets a higher voltage than pin 2 making the output of IC1 high. Meanwhile, capacitor C4 charges to full voltage and keeps pin 3 high for a few minutes even after T1 is non-conducting. Resistor R3 provides discharge path for capacitor C4, which decides the time period for which the output of comparator IC1 should remain high.

The high output of IC1 energises relay RL1 through relay-driver transistor T2. Thus the load, i.e., security lamps, turn on for three to four minutes. LED2 glows to indicate activation of the relay as well as switching 'on' of the security lights. Connect a single-pole, single-throw 'on'/'off' switch (MS) to activate the security lamps manually when required.

Zener diode ZD1 provides 5.1V DC for safe operation of the IR receiver and associated circuit. Power for the circuit is derived from a step-down transformer (X1) and a bridge rectifier comprising diodes D1 through D4. Smoothing capacitor C1 removes ripples, if any, from the power supply.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Drill holes on the front panel for mounting the IR sensor and LEDs. Connect the master switch between the normally-open (N/O) contact and pole of relay RL1 so that the master switch can be used when needed. The relay contacts rating should be more than 4A. Mount the unit near the master switch using minimal wiring. ●

USB POWER SOCKET



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Today, almost all computers contain logic blocks for implementing a USB port. A USB port, in practice, is capable of delivering more than 100 mA of continuous current at 5V to the peripherals that are connected to the bus. So a USB port can be used, without any trouble, for powering 5V DC operated tiny electronic gadgets.

Nowadays, many handheld devices (for instance, portable reading lamps) utilise this facility of the USB port to recharge their built-in battery pack with the help of an internal

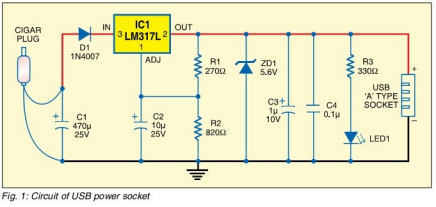


Fig. 1: Circuit of USB power socket